

Question	Answer	Marks	Guidance
3(a)	(objects are at) same temperature	B1	Do not allow 'constant' for 'same'
	(if in thermal contact) no <u>net transfer</u> of thermal energy (between them)	B1	Allow 'heat' for 'thermal energy'. Allow 'overall' or 'resultant' for 'net'. Allow 'exchange' or 'flow between them' for 'transfer'. Do not allow 'change' for 'transfer'.
3(b)(i)	thermistor absorbs thermal energy (to raise its temperature)	B1	Allow 'heat' for 'thermal energy'.
	thermal energy is transferred from the water, which cools	B1	
3(b)(ii)	$Q = mc\Delta\theta$	C1	
	$63 \times 4.2 \times (78.0 - 76.8) = 8.5 \times c \times (76.8 - 25.0)$	C1	
	specific heat capacity = $0.72 \text{ J g}^{-1} \text{ K}^{-1}$	A1	Correct to at least 2 significant figures. AFC. (0.721)
3(c)	use a thermocouple	B1	Allow 'use a thermistor that is <u>smaller</u> / has lower mass'.
	smaller mass so will remove less thermal energy (from water)	B1	Allow 'energy' or 'heat' for 'thermal energy'. Allow 'heat capacity' for 'mass'
	or		
	warm the thermistor (to approximately 78°C) before immersing it in the water	(B1)	For 2 / 2, both B1 marks must be from same method.
	smaller temperature change to reach equilibrium so will remove less thermal energy (from the water)	(B1)	